

Engineering Economics Trading Bot Project

Introduction

This project develops a robust trading bot capable of both backtesting and live trading using real-time data streams. Leveraging advanced technical indicators such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Exponential Moving Average (EMA), the bot dynamically generates buy and sell signals for six cryptocurrencies: ETHUSDT, BTCUSDT, AVAXUSDT, SOLUSDT, RENDERUSDT, and FETUSDT. The system integrates WebSocket connections for high-frequency real-time data processing and allows efficient parameter customization for strategy optimization.

Backtesting Strategy

The backtesting phase simulates past market conditions to evaluate the algorithm's potential performance. Using the **RunBacktest** method, the bot processes historical data to generate buy and sell signals based on RSI, MACD, and EMA indicators. This functionality applies across multiple intervals, including 15 minutes (15m), 1 hour (1h), 4 hours (4h), and 1 day (1d).

Methodology

Data Fetching

- Historical candlestick data is fetched using Binance's API via the `_binanceService.GetHistoricalData` method.
- The data includes open, high, low, close, and volume values for each interval.

Indicator Calculations

- **RSI**: Measures the relative strength of price movements to identify overbought and oversold conditions.
- **MACD**: Highlights momentum changes by comparing the short-term and long-term exponential moving averages.
- **EMA**: Provides a smoothed representation of price trends to reduce noise.

Signal Logic

- **Buy Signal**:

- Triggered when the RSI is below the buy threshold, indicating oversold conditions.
- MACD is slightly above its signal line ($\text{MACD} > \text{Signal} - 0.1$), reflecting bullish momentum.
- The close price is above 85% of the EMA value, signifying a strong upward trend.
- **Sell Signal:**
 - Triggered when RSI exceeds the sell threshold, indicating overbought conditions.
 - MACD is slightly below its signal line ($\text{MACD} < \text{Signal} + 0.1$), reflecting bearish momentum.

Trade Simulation

- Trades are executed virtually, calculating the position size based on the available capital.
- Trades are logged with detailed information, including type (buy/sell), date, price, quantity, and resulting profit or loss.

Output and Visualization

- **Final Results:** The net profit or loss, trade history, and final capital are displayed on the front-end.
- **Customization:** Users can manually input RSI, MACD, and EMA values on the frontend for parameter experimentation.
- **Interval Comparison:** Backtesting results for 15m, 1h, 4h, and 1d intervals provide insights into the algorithm's adaptability for different trading strategies.

Live Trading Strategy

The live trading phase employs the **RunBacktestOnline** method to process real-time data streams and make buy/sell decisions dynamically. The system uses a 1-minute (1m) interval for live trading and connects to Binance's live trading streams via WebSocket.

Methodology

WebSocket Integration

- **Data Processing:** Real-time trade events are fetched through WebSocket and aggregated into candlestick objects.
- **Interval Handling:** A timer ensures trades are grouped into 1-minute intervals.

- **Indicator Updates:** RSI, MACD, and EMA values are recalculated dynamically as data streams in.

Signal Logic

- The live trading logic mirrors the backtesting logic, using RSI, MACD, and EMA indicators to generate buy and sell signals in real time.

Execution and Feedback

- **Real-Time Execution:** Trades are executed instantly based on the generated signals, with each trade logged for transparency.
- **Frontend Integration:** SignalR (provided by .NET 8) streams trade data and updated capital balances to the frontend for live visualization.
- **Scalability:** The system processes thousands of real-time trades per second, ensuring robust performance during high-volatility periods.

Live Trading Experiment

In the live trading experiment, six different cryptocurrencies, including ETHUSDT and BTCUSDT, were tested with varying RSI thresholds to evaluate real-time trading feasibility. It was observed that for high-liquidity coins like ETHUSDT and BTCUSDT, long waiting periods were required for the conditions to be met, often exceeding several hours due to the slower price movements that align with RSI and other indicator thresholds.

Conversely, for FETUSDT, the conditions were met more quickly, allowing a trade to execute within a relatively short waiting time of approximately 30 minutes. This highlights the importance of selecting assets with higher price volatility when aiming for frequent trade opportunities in live trading scenarios.

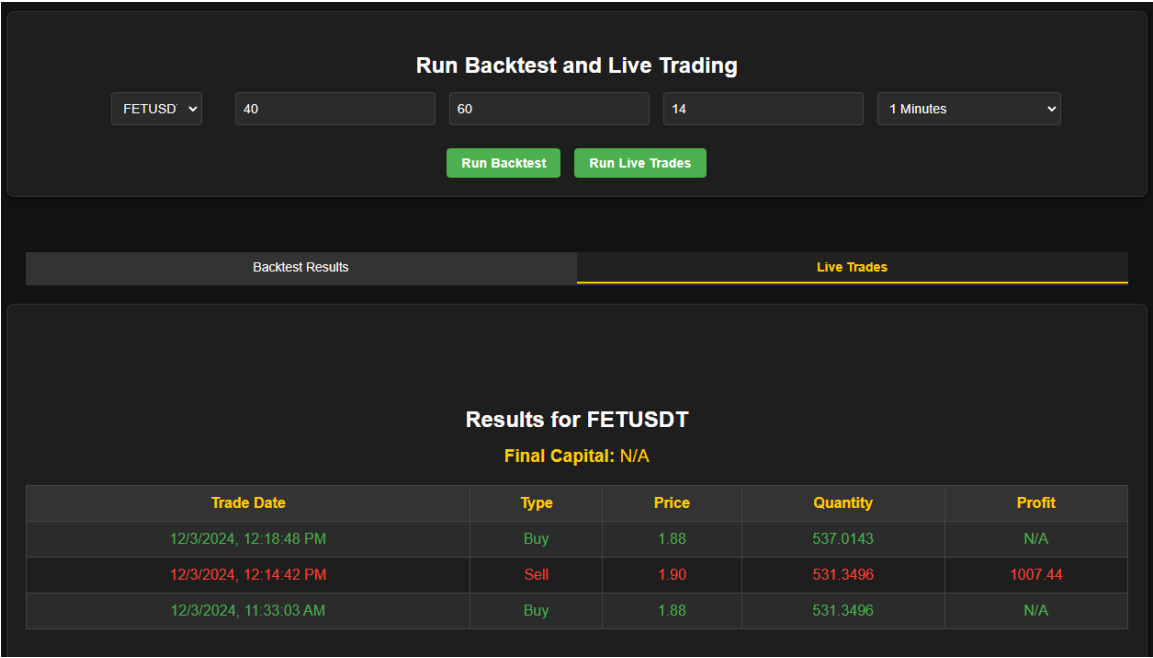
In the attached screenshot, you can observe the results of live trading for FETUSDT with a one-minute interval. The bot successfully executed both a "Buy" and a "Sell" transaction within the specified timeframe, achieving a final capital of **1007.44 USDT** after the sell trade. This demonstrates the bot's ability to adapt and execute trades efficiently when conditions are favorable.

Key observations:

- **Coin Selection:** FETUSDT provided a faster trading opportunity compared to more stable coins like ETHUSDT and BTCUSDT.
- **Execution:** The bot processed and completed trades in alignment with the predefined conditions in under an hour.

- **Efficiency:** The strategy is particularly effective for smaller, more volatile coins where trading conditions are met more frequently.

This experiment underscores the importance of tailoring strategies and coin selections to market conditions to optimize trade execution and efficiency.



Output and Visualization

- **Trade Monitoring:** Users can view real-time buy/sell transactions, updated balances, and trade histories on the front-end.
- **Graphical Display:** Each coin’s candlestick data and live price are presented in charts and tables. Graphs for RSI, MACD, and EMA values allow users to track trends and market conditions visually.

Validation and Comparison

The live trading results are compared against backtesting simulations to validate the algorithm’s robustness under real-world conditions. Discrepancies between the two are used to fine-tune parameters for better risk management and profitability.

Key Metrics

- **Backtesting Outputs:** Net profit/loss, final capital, and trade history for various intervals.
- **Live Trading Outputs:** Real-time capital updates, trade frequency, and execution metrics.

By continuously refining the system based on these comparisons, the trading bot ensures optimal performance across both simulated and live environments.

Frontend Features

The frontend enhances user experience by providing interactive and informative tools for monitoring and customization:

- **Custom Parameter Inputs:** Users can input RSI, MACD, and EMA thresholds to experiment with different strategies.
- **Live Data Display:** Graphs and tables show up-to-date candlestick data, current prices, and technical indicator values for all supported coins.
- **Real-Time Visualization:** SignalR integration ensures real-time updates for executed trades and capital changes.

Graphical Representation

The bot includes comprehensive graphical tools for visualizing trading performance:

- **Candlestick Charts:** Real-time and historical price trends.
- **Technical Indicators:** Visual representation of RSI, MACD, and EMA trends.
- **Performance Metrics:** Graphs showing profit/loss over time, trade frequencies, and other critical statistics.

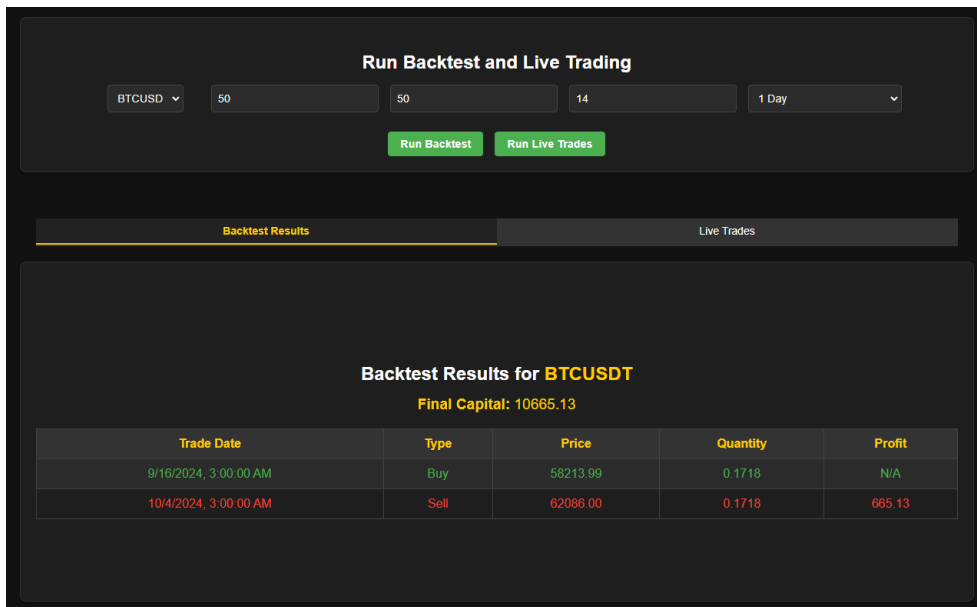
These visual elements provide users with deeper insights into the bot's operation and the market dynamics, enhancing decision-making and strategy optimization.

Backtest Results

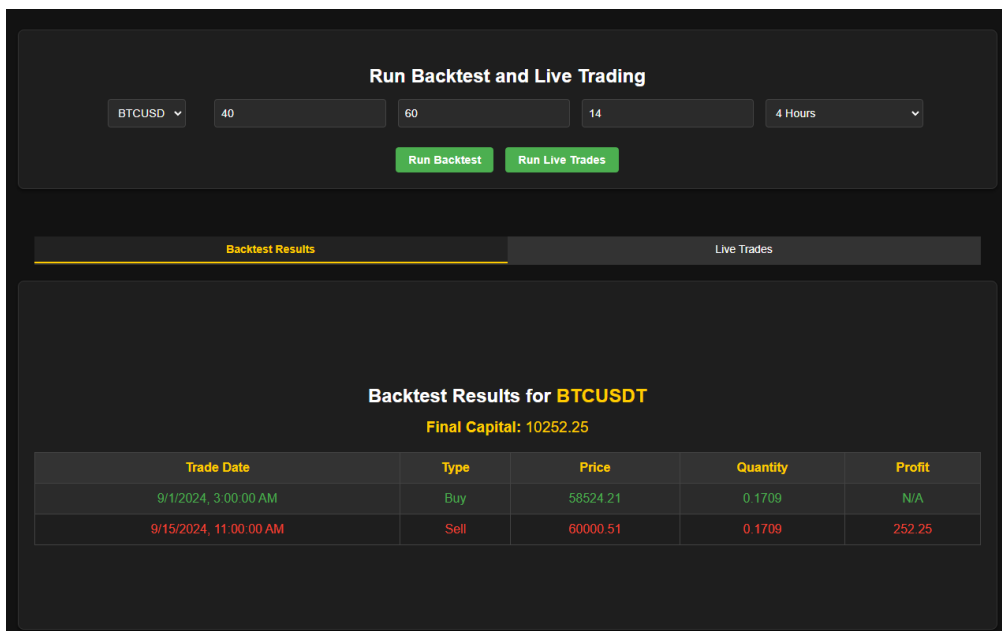
In this section, we present the backtest results for various cryptocurrencies and intervals using different RSI, MACD, and EMA thresholds. The objective of this analysis is to evaluate the algorithm's performance in terms of profitability and risk management. Each result is accompanied by a detailed breakdown of trades and the corresponding profit or loss.

1. BTCUSDT

BTCUSDT, 1d Interval: Backtest results for BTCUSDT on a 1-day interval reveal a solid performance, achieving a net final capital of 10665.13 with RSI thresholds of 50 (buy) and 50 (sell).



BTCUSDT, 4h Interval: Backtest results for BTCUSDT on a 4-hour interval reveal a solid performance, achieving a net final capital of 10252.25 with RSI thresholds of 40 (buy) and 60 (sell).



BTCUSDT, 1h Interval: Backtest results for BTCUSDT on a 1-hour interval reveal a solid performance, achieving a net final capital of N/A with RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

BTCUSD

40

60

14

1 Hour

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for BTCUSDT

Final Capital: N/A

Trade Date	Type	Price	Quantity	Profit
8/4/2024, 10:00:00 AM	Buy	60373.99	0.1656	N/A
8/9/2024, 9:00:00 AM	Sell	60890.01	0.1656	85.47
8/12/2024, 4:00:00 PM	Buy	58157.55	0.1734	N/A
8/14/2024, 1:00:00 PM	Sell	61100.00	0.1734	595.74
8/15/2024, 9:00:00 PM	Buy	57872.82	0.1831	N/A
8/18/2024, 3:00:00 AM	Sell	59678.93	0.1831	926.41
8/28/2024, 4:00:00 PM	Buy	59334.00	0.1842	N/A
9/10/2024, 8:00:00 AM	Sell	56945.45	0.1842	486.56
9/17/2024, 1:00:00 AM	Buy	57970.01	0.1809	N/A

BTCUSDT, 15m Interval: Backtest results for BTCUSDT on a 15-minute interval reveal a solid performance, achieving a net final capital of 10665.13 with RSI thresholds of 50 (buy) and 50 (sell).

Run Backtest and Live Trading

BTCUSD

40

60

14

15 Minutes

Run Backtest

Run Live Trades

Backtest Results

Live Trades

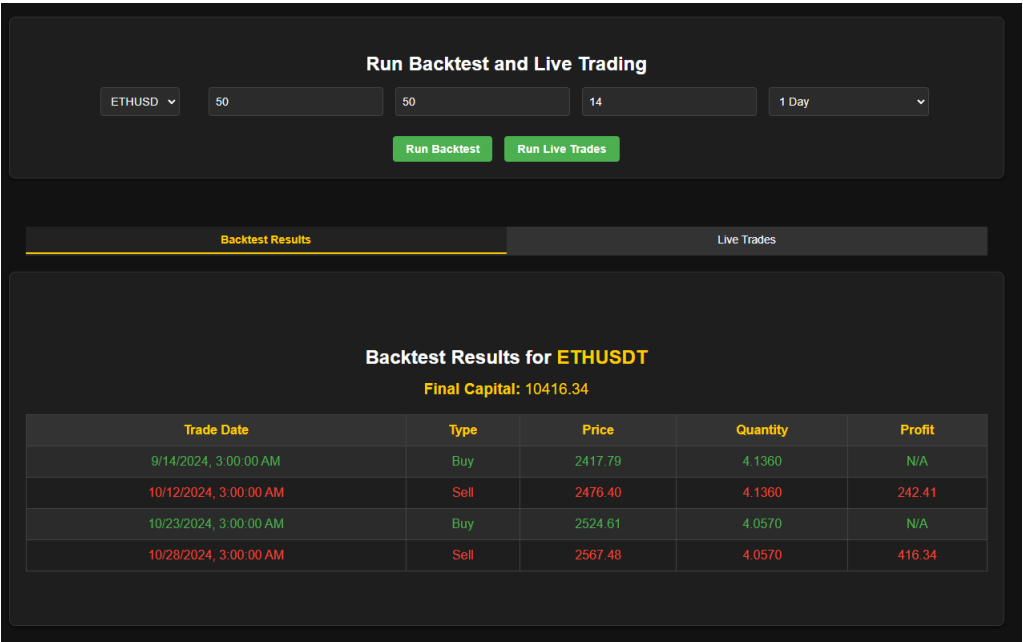
Backtest Results for BTCUSDT

Final Capital: N/A

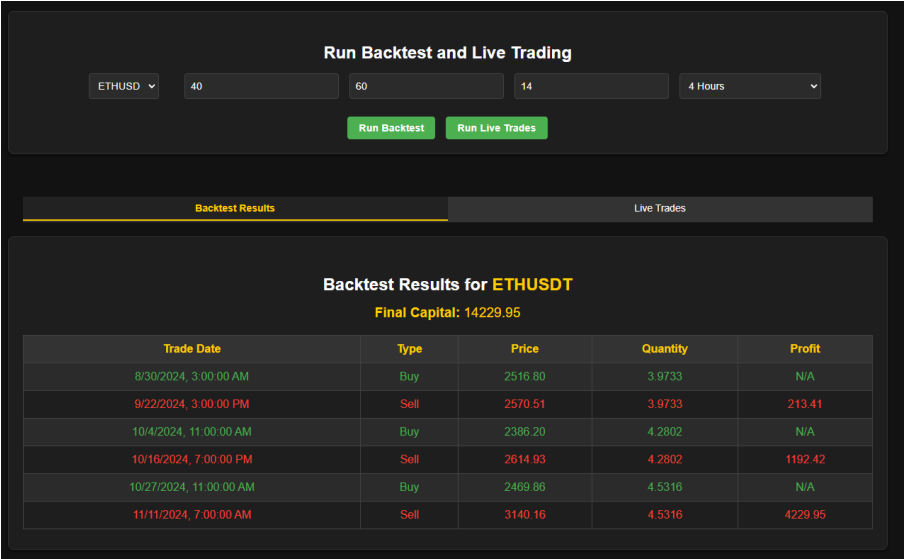
Trade Date	Type	Price	Quantity	Profit
8/3/2024, 11:45:00 PM	Buy	60395.97	0.1656	N/A
8/6/2024, 6:15:00 AM	Sell	55751.99	0.1656	-768.92
8/7/2024, 11:30:00 PM	Buy	55016.50	0.1678	N/A
8/8/2024, 9:15:00 AM	Sell	57154.45	0.1678	-410.20
8/11/2024, 8:00:00 PM	Buy	60154.49	0.1594	N/A
8/13/2024, 10:15:00 PM	Sell	60789.07	0.1594	-309.20
8/14/2024, 8:45:00 PM	Buy	59168.36	0.1638	N/A
8/16/2024, 12:15:00 PM	Sell	58508.28	0.1638	-417.31
8/19/2024, 5:30:00 AM	Buy	58567.60	0.1636	N/A
8/20/2024, 6:45:00 AM	Sell	60619.18	0.1636	-81.63

2. ETHUSDT

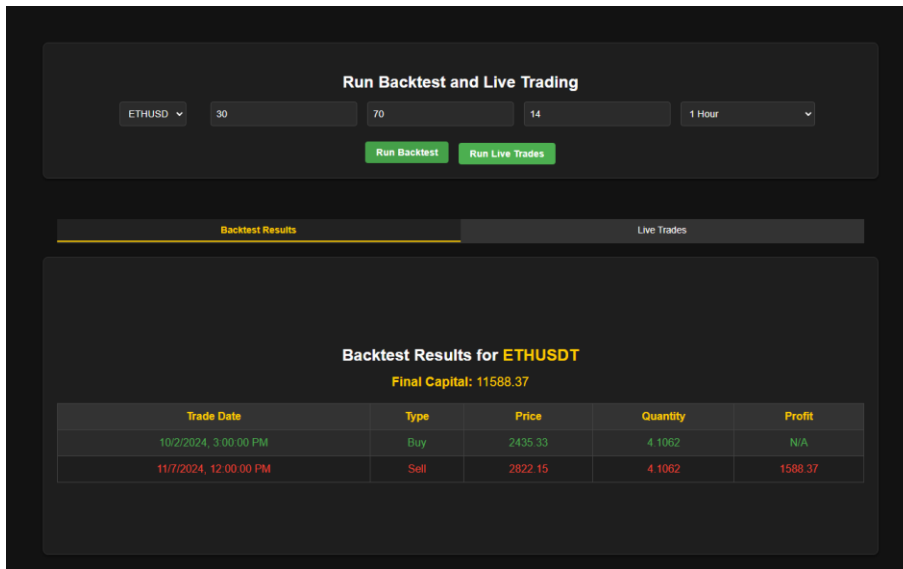
ETHUSDT, 1d Interval: Backtest results for **ETHUSDT** on a 1-day interval reveal a solid performance, achieving a net final capital of 10416.34 with RSI thresholds of 50 (buy) and 50 (sell).



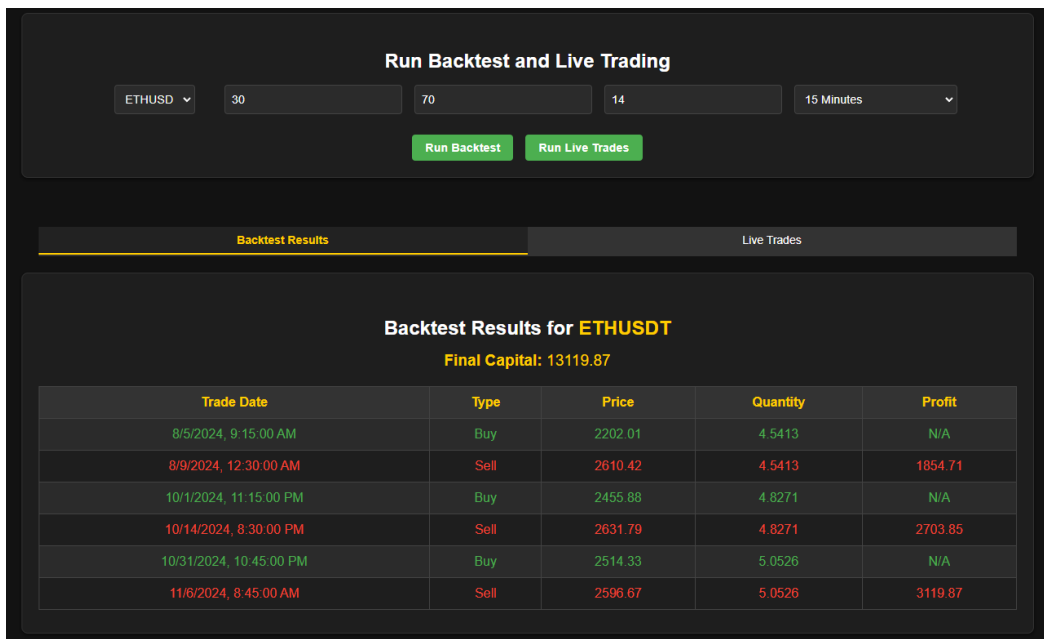
ETHUSDT, 4h Interval: Backtest results for **ETHUSDT** on a 4-hour interval reveal a solid performance, achieving a net final capital of 14229.95 with RSI thresholds of 40 (buy) and 60 (sell).



ETHUSDT, 1h Interval: Backtest results for **ETHUSDT** on a 1-hour interval reveal a solid performance, achieving a net final capital of 11588.37 with RSI thresholds of 30 (buy) and 70 (sell).



ETHUSD, 15m Interval: Backtest results for **ETHUSD** on a 15-minute interval reveal a solid performance, achieving a net final capital of 10416.34 with RSI thresholds of 50 (buy) and 50 (sell).



3. AVAXUSD

AVAXUSDT, 1d Interval: Backtest results for AVAXUSDT on a 1-day interval display steady performance, achieving a final capital of 10174.28 with RSI thresholds of 50 (buy) and 50 (sell).

Run Backtest and Live Trading

AVAXUSDT

50

50

14

1 Day

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for AVAXUSDT

Final Capital: 10174.28

Trade Date	Type	Price	Quantity	Profit
10/1/2024, 3:00:00 AM	Buy	25.82	387.2967	N/A
10/4/2024, 3:00:00 AM	Sell	26.27	387.2967	174.28

AVAXUSDT, 4h Interval: Backtest results for AVAXUSDT on a 4-hour interval highlight significant profitability, achieving a final capital of 14173.23 with RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

AVAXUSDT

40

60

14

4 Hours

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for AVAXUSDT

Final Capital: 14173.23

Trade Date	Type	Price	Quantity	Profit
8/11/2024, 11:00:00 PM	Buy	20.40	490.1961	N/A
8/18/2024, 7:00:00 PM	Sell	21.35	490.1961	465.69
8/30/2024, 3:00:00 PM	Buy	22.38	467.6357	N/A
9/10/2024, 7:00:00 AM	Sell	23.80	467.6357	1129.73
9/18/2024, 11:00:00 AM	Buy	23.25	478.6980	N/A
9/19/2024, 3:00:00 AM	Sell	25.82	478.6980	2264.24
10/4/2024, 3:00:00 AM	Buy	24.92	492.1446	N/A
10/11/2024, 7:00:00 PM	Sell	27.32	492.1446	3445.39
10/23/2024, 3:00:00 AM	Buy	27.34	491.7846	N/A
11/9/2024, 3:00:00 PM	Sell	28.82	491.7846	4173.23

AVAXUSDT, 1h Interval: Backtest results for AVAXUSDT on a 1-hour interval reveal dynamic performance, focusing on shorter interval trends under RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

AVAXUSDT

40

60

14

1 Hour

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for AVAXUSDT

Final Capital: 9790.33

Trade Date	Type	Price	Quantity	Profit
8/4/2024, 10:00:00 AM	Buy	22.59	442.6737	N/A
8/6/2024, 5:00:00 PM	Sell	21.04	442.6737	-686.14
8/7/2024, 5:00:00 PM	Buy	20.37	457.2340	N/A
8/8/2024, 5:00:00 PM	Sell	21.39	457.2340	-219.77
8/11/2024, 4:00:00 PM	Buy	21.52	454.4719	N/A
8/12/2024, 2:00:00 PM	Sell	21.53	454.4719	-215.22
8/15/2024, 12:00:00 AM	Buy	20.94	467.2769	N/A
8/18/2024, 2:00:00 AM	Sell	20.70	467.2769	-327.37
8/19/2024, 8:00:00 AM	Buy	20.43	468.8625	N/A
8/20/2024, 1:00:00 AM	Sell	21.09	468.8625	-111.69

AVAXUSDT, 15m Interval: Backtest results for AVAXUSDT on a 15-minute interval highlight significant profitability, achieving a final capital of 10631.23 with RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

AVAXUSDT

40

60

14

15 Minutes

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for AVAXUSDT

Final Capital: 10631.23

Trade Date	Type	Price	Quantity	Profit
8/3/2024, 11:30:00 AM	Buy	23.56	424.4482	N/A
8/3/2024, 1:30:00 PM	Sell	23.87	424.4482	131.58
8/3/2024, 6:30:00 PM	Buy	23.53	430.5813	N/A
8/5/2024, 8:00:00 PM	Sell	19.81	430.5813	-1470.18
8/6/2024, 4:00:00 PM	Buy	20.26	421.0176	N/A
8/6/2024, 5:00:00 PM	Sell	20.92	421.0176	-1192.31
8/7/2024, 3:00:00 AM	Buy	20.70	425.4922	N/A
8/7/2024, 7:00:00 AM	Sell	21.39	425.4922	-898.72
8/7/2024, 2:15:00 PM	Buy	21.03	432.7759	N/A

4. FETUSDT

FETUSDT, 1d Interval: Backtest results for **FETUSDT** on a 1-day interval display steady performance, achieving a final capital of 15181.24 with RSI thresholds of 50 (buy) and 50 (sell).

Run Backtest and Live Trading

FETUSD

50

50

14

1 Day

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for FETUSDT

Final Capital: 15181.24

Trade Date	Type	Price	Quantity	Profit
9/5/2024, 3:00:00 AM	Buy	1.08	9259.2593	N/A
9/8/2024, 3:00:00 AM	Sell	1.10	9259.2593	194.44
10/2/2024, 3:00:00 AM	Buy	1.42	7174.1340	N/A
10/4/2024, 3:00:00 AM	Sell	1.47	7174.1340	517.28
10/5/2024, 3:00:00 AM	Buy	1.41	7480.2848	N/A
10/6/2024, 3:00:00 AM	Sell	1.47	7480.2848	1025.94
10/8/2024, 3:00:00 AM	Buy	1.42	7748.3766	N/A
10/12/2024, 3:00:00 AM	Sell	1.48	7748.3766	1444.35

FETUSDT, 4h Interval: Backtest results for **FETUSDT** on a 4-hour interval highlight significant profitability, achieving a final capital of 13913.56 with RSI thresholds of 50 (buy) and 50 (sell).

Run Backtest and Live Trading

FETUSD

50

50

14

4 Hours

Run Backtest

Run Live Trades

Backtest Results

Live Trades

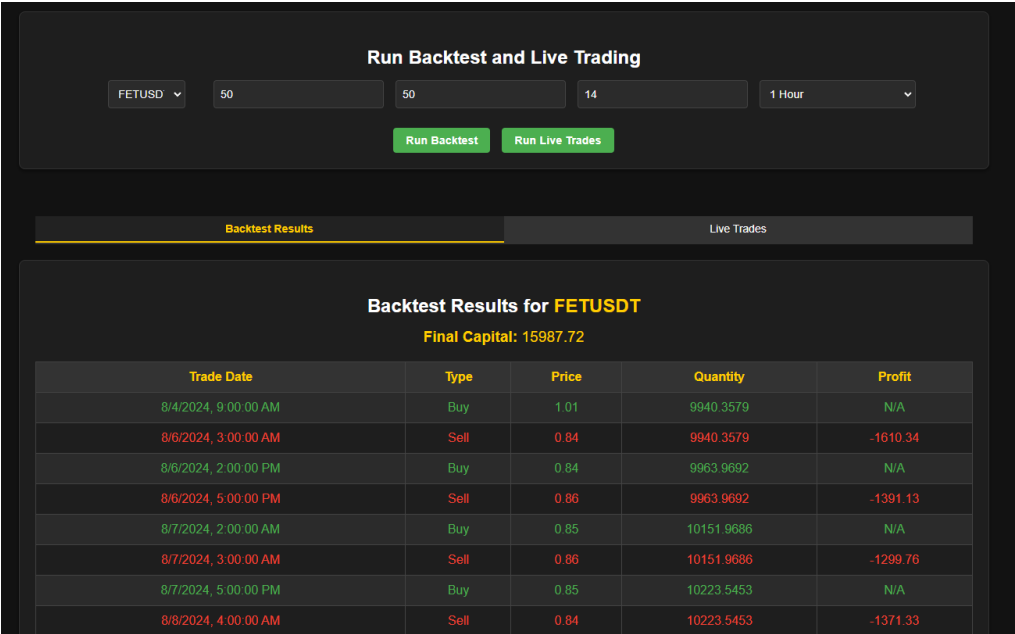
Backtest Results for FETUSDT

Final Capital: 13913.58

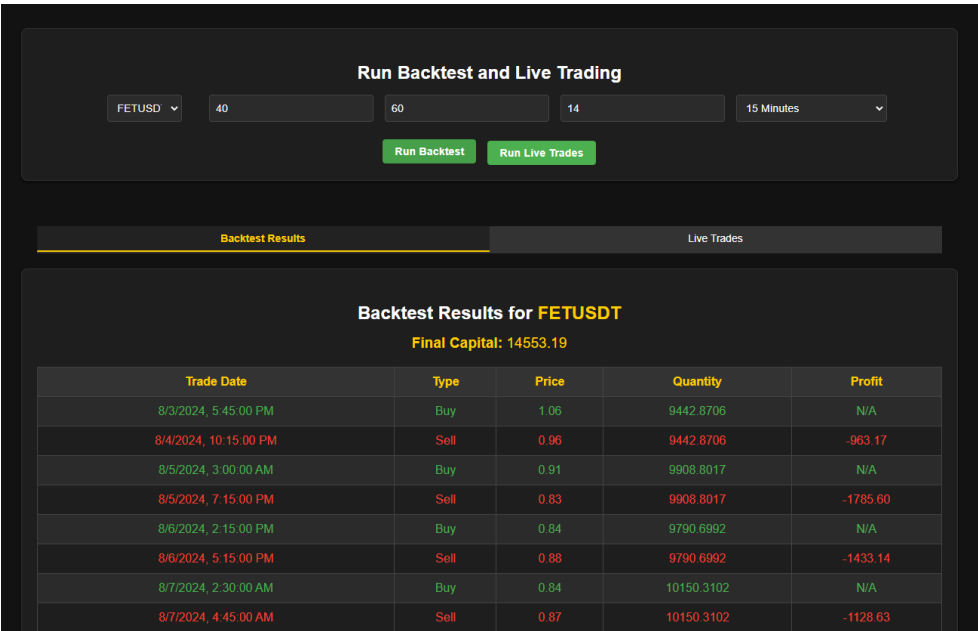
Trade Date	Type	Price	Quantity	Profit
8/8/2024, 3:00:00 PM	Buy	0.85	11737.0892	N/A
8/8/2024, 11:00:00 PM	Sell	0.89	11737.0892	410.80
8/9/2024, 7:00:00 AM	Buy	0.86	12133.7973	N/A
8/10/2024, 11:00:00 PM	Sell	0.90	12133.7973	908.28
8/11/2024, 3:00:00 PM	Buy	0.86	12684.0509	N/A
8/12/2024, 11:00:00 AM	Sell	0.89	12684.0509	1314.17
8/12/2024, 7:00:00 PM	Buy	0.85	13279.5463	N/A
8/13/2024, 3:00:00 PM	Sell	0.86	13279.5463	1460.25

FETUSDT, 1h Interval: Backtest results for **FETUSDT** on a 1-hour interval highlight significant profitability, achieving a final capital of 15967.72 with RSI thresholds of 50 (buy)

and 50 (sell).

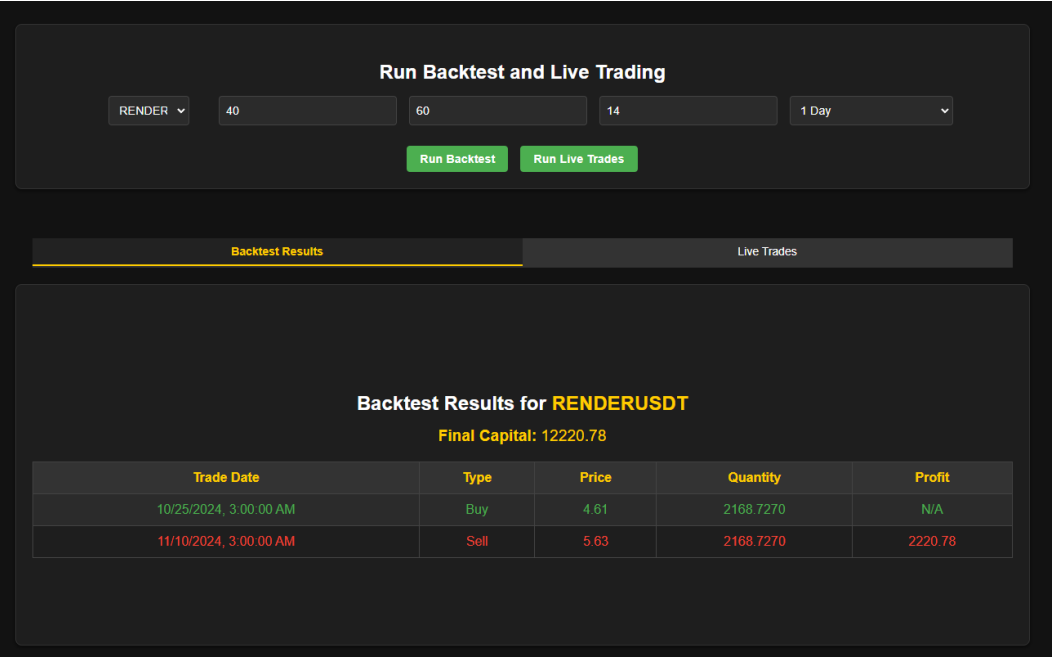


FETUSD, 15m Interval: "Backtest results for **FETUSD** on a 15-minute interval highlight significant profitability, achieving a final capital of 14553.19 with RSI thresholds of 40 (buy) and 60 (sell)."

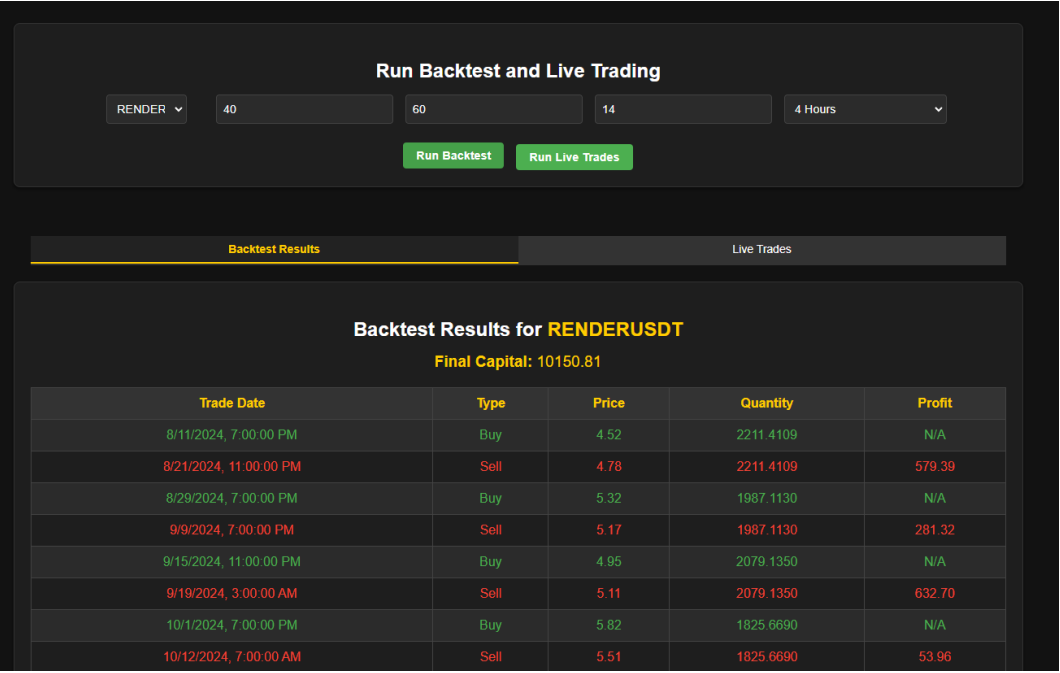


5. RENDERUSD

RENDERUSD, 1d Interval: Backtest results for RENDERUSD on a 1-day interval demonstrate strong performance, achieving a final capital of 12220.78 with RSI thresholds of 40 (buy) and 60 (sell).



RENDERUSDT, 4h Interval: Backtest results for RENDERUSDT on a 4-hour interval reveal a net capital of 10150.81, using RSI thresholds of 40 (buy) and 60 (sell).



RENDERUSDT, 1h Interval: Backtest results for RENDERUSDT on a 1-hour interval indicate profitable trades, with a final capital of N/A using RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

RENDER ▾

40

60

14

1 Hour ▾

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for RENDERUSDT

Final Capital: N/A

Trade Date	Type	Price	Quantity	Profit
8/4/2024, 5:00:00 PM	Buy	4.62	2163.5656	N/A
8/6/2024, 8:00:00 AM	Sell	4.75	2163.5656	274.77
8/7/2024, 5:00:00 PM	Buy	4.56	2253.7339	N/A
8/8/2024, 5:00:00 PM	Sell	4.78	2253.7339	777.36
8/10/2024, 8:00:00 AM	Buy	4.76	2262.7242	N/A
8/11/2024, 12:00:00 AM	Sell	4.92	2262.7242	1143.92
8/11/2024, 3:00:00 PM	Buy	4.78	2330.8757	N/A
8/12/2024, 2:00:00 PM	Sell	4.75	2330.8757	1076.32

RENDERUSDT, 15m Interval: Backtest results for RENDERUSDT on a 15-minute interval highlight effective trading strategies, achieving a final capital of 22985.39 with RSI thresholds of 50 (buy) and 50 (sell).

Run Backtest and Live Trading

RENDER ▾

50

50

14

15 Minutes ▾

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for RENDERUSDT

Final Capital: 22985.39

Trade Date	Type	Price	Quantity	Profit
8/3/2024, 7:45:00 AM	Buy	4.91	2037.4898	N/A
8/3/2024, 8:00:00 AM	Sell	4.95	2037.4898	85.57
8/3/2024, 6:15:00 PM	Buy	5.14	1961.4108	N/A
8/3/2024, 11:30:00 PM	Sell	4.95	1961.4108	-291.02
8/4/2024, 1:30:00 AM	Buy	4.95	1960.6187	N/A
8/4/2024, 2:45:00 AM	Sell	4.95	1960.6187	-294.94
8/4/2024, 3:00:00 AM	Buy	4.92	1973.7772	N/A
8/4/2024, 3:30:00 AM	Sell	4.95	1973.7772	-235.72

6. SOLUSDT

SOLUSDT, 1d Interval: Backtest results for SOLUSDT on a 1-day interval reflect profitability, achieving a final capital of 10766.63 with RSI thresholds of 50 (buy) and 50

(sell)

Run Backtest and Live Trading

SOLUSD

50

50

14

1 Day

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for SOLUSD

Final Capital: 10766.63

Trade Date	Type	Price	Quantity	Profit
9/12/2024, 3:00:00 AM	Buy	136.05	73.5024	N/A
10/6/2024, 3:00:00 AM	Sell	146.48	73.5024	766.63

SOLUSD, 4h Interval: Backtest results for SOLUSD on a 4-hour interval deliver a solid outcome, with a final capital of 13576.49 using RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

SOLUSD

40

60

14

4 Hours

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for SOLUSD

Final Capital: 13576.49

Trade Date	Type	Price	Quantity	Profit
8/31/2024, 3:00:00 PM	Buy	136.63	73.1904	N/A
9/22/2024, 3:00:00 AM	Sell	148.09	73.1904	838.76
11/4/2024, 7:00:00 AM	Buy	161.41	67.1505	N/A
11/9/2024, 11:00:00 AM	Sell	202.18	67.1505	3576.49

SOLUSDT, 1h Interval: Backtest results for SOLUSDT on a 1-hour interval indicate mixed results, with a final capital of N/A using RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

SOLUSD

40

60

14

1 Hour

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for SOLUSDT

Final Capital: N/A

Trade Date	Type	Price	Quantity	Profit
8/4/2024, 9:00:00 AM	Buy	143.71	69.5846	N/A
8/7/2024, 4:00:00 AM	Sell	149.13	69.5846	377.15
8/12/2024, 9:00:00 AM	Buy	143.46	72.3348	N/A
8/18/2024, 12:00:00 PM	Sell	142.99	72.3348	343.15
8/21/2024, 12:00:00 PM	Buy	141.52	73.0861	N/A
8/23/2024, 3:00:00 AM	Sell	144.40	73.0861	553.64
8/27/2024, 11:00:00 PM	Buy	154.02	68.5212	N/A
9/5/2024, 4:00:00 PM	Sell	134.54	68.5212	-781.15

SOLUSDT, 15m Interval: Backtest results for SOLUSDT on a 15-minute interval emphasize variability, achieving a final capital of N/A with RSI thresholds of 40 (buy) and 60 (sell).

Run Backtest and Live Trading

SOLUSD

40

60

14

15 Minutes

Run Backtest

Run Live Trades

Backtest Results

Live Trades

Backtest Results for SOLUSDT

Final Capital: N/A

Trade Date	Type	Price	Quantity	Profit
8/3/2024, 11:15:00 PM	Buy	142.06	70.3928	N/A
8/6/2024, 12:00:00 AM	Sell	133.25	70.3928	-620.16
8/7/2024, 11:30:00 PM	Buy	144.42	64.9483	N/A
8/8/2024, 9:15:00 AM	Sell	153.99	64.9483	1.40
8/9/2024, 5:15:00 PM	Buy	154.13	64.8893	N/A
8/10/2024, 9:00:00 PM	Sell	155.04	64.8893	60.44
8/11/2024, 7:15:00 PM	Buy	149.03	67.5062	N/A
8/12/2024, 4:00:00 PM	Sell	148.98	67.5062	57.07

